



LENDINGCLUB: LOAN DEFAULT PREDICTION MODEL

Group 5

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EXECUTIVE SUMMARY

This study develops and evaluates a loan default prediction model for LendingClub, with the objective of assessing default risk at the time of loan origination.

After standardising the target outcome and restricting predictors to information observable at application time, a final modelling sample of 517,959 historical loans was retained, labelled as default versus fully paid. Twenty-six economically interpretable features were selected, covering loan contract structure, borrower repayment capacity, and credit history indicators. Data preparation included complete case filtering for missingness, controlled capping and transformation of extreme values, and tailored encoding of categorical variables.

The data was partitioned into training, validation, and test sets, and three model classes were estimated: Logistic Regression, LightGBM, and tuned XGBoost. All models achieved test ROC AUC close to 0.70, indicating meaningful ability to rank applicants by default risk. Tuned XGBoost delivered the strongest overall balance, with ROC AUC 0.708, precision recall AUC 0.306, and recall approximately 0.67, implying that roughly two thirds of future defaulters can be identified prior to funding.

On this basis, the tuned XGBoost model is selected as the primary scoring engine for deployment. Its predicted probability of default is then mapped to a four-tier decision policy spanning automatic approval, manual review, risk adjusted pricing, and rejection. Logistic Regression is retained as an interpretability layer for regulatory communication and governance.

1. Introduction

1.1. Business context

The rapid growth of financial technologies (FinTech) has reshaped the global credit landscape, driving the expansion of the **alternative lending market**, a core element of today's digital financial ecosystem. Unlike traditional banks constrained by regulation and conservative lending policies, alternative lenders leverage big data, artificial intelligence, and cloud systems to connect borrowers and investors efficiently. Valued at USD 35 billion in 2024, the sector is projected to reach USD 62 billion by 2033 (Business Research Insights, 2024), reflecting rising demand for fast, accessible credit and a shift toward technology-driven finance.

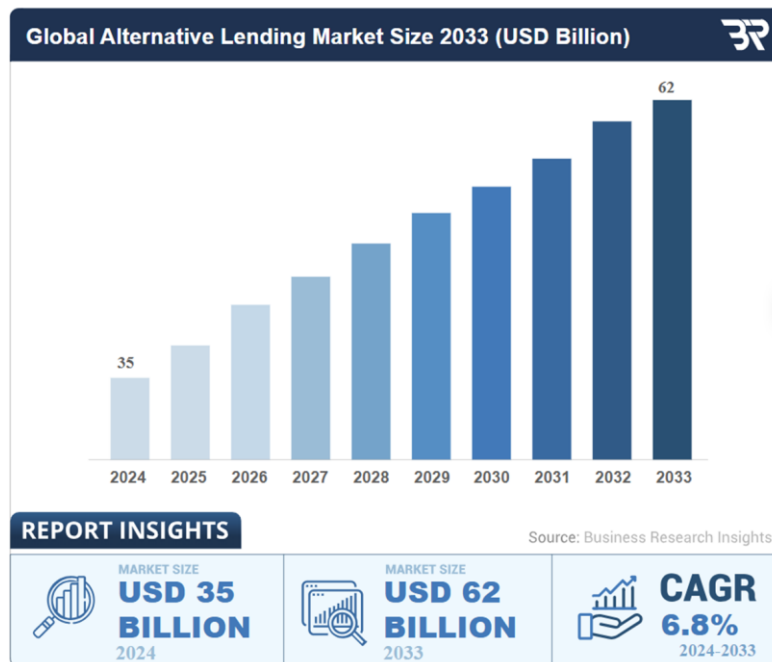


Figure 1: Global alternative lending market size 2033 (Business Research Insights, 2025)

Since the 2008 Global Financial Crisis, alternative lending models such as peer-to-peer (P2P) lending have emerged as key players in credit provision worldwide. The crisis triggered a tightening of bank lending standards, which made access to credit significantly harder for individuals and small-to-medium enterprises (SMEs). Meanwhile, persistent low interest rates in developed economies encouraged investors to seek higher-yield opportunities outside conventional banking products. Within this context, **LendingClub**,

founded in 2007 in San Francisco, stands out as a pioneer and remains one of the most studied examples of FinTech-based credit innovation. Initially launched as a pure P2P platform, LendingClub enabled retail investors to fund consumer loans directly. Over time, its model evolved into a hybrid marketplace banking system, incorporating both institutional investors and bank partnerships after the acquisition of Radius Bank in 2020.

1.2. Business problem

Although LendingClub has advanced technologically, it continues to face the core challenge of the alternative lending sector - **credit default risk**. Because loans are largely unsecured and borrower information is limited, default remains a primary concern for both investors and platforms. This project addresses the **inefficiency of current credit scoring systems**, as LendingClub's internal grades and traditional FICO scores show only moderate ability to distinguish between creditworthy and high-risk borrowers. Information asymmetry and class imbalance further weaken predictive performance, leading many models to overfit non-default cases while misclassifying risky borrowers. These limitations reduce predictive reliability, expose investors to potential losses, and threaten the long-term credibility of alternative lending platforms.

1.3. Motivation

Motivation for this project is both practical and academic:

From a business perspective, more **accurate default prediction models** can reduce investor losses, improve portfolio returns, and enhance platform transparency, thereby increasing user confidence and supporting market stability. Even small improvements in predictive performance, such as a one or two-percentage-point increase in the Area Under the Curve (AUC) - can translate into substantial savings across large loan portfolios.

From an academic standpoint, this research contributes to understanding how machine learning (ML) techniques can be effectively applied in real-world financial settings to overcome data imbalance, feature noise, and model interpretability challenges. As the alternative lending market continues to expand, developing reliable and transparent risk assessment models is essential not only for investor protection but also for the long-term sustainability of digital credit ecosystems.

1.4. Contribution

This study makes several key contributions:

Problem framing: The project situates LendingClub within the broader alternative lending market, offering a modern business context that links FinTech innovation with risk management and investor protection.

Data preparation and feature engineering: Using LendingClub's public dataset, this study conducts comprehensive data cleaning, transformation, and feature selection, ensuring only pre-funding variables (those available before loan issuance) are included, thus maintaining real-world relevance.

Model comparison and evaluation: Several ML algorithms - including Logistic Regression, LightGBM and XGBoost are implemented and benchmarked using consistent cross-validation. The models are evaluated through robust metrics such as Accuracy, ROC-AUC, PR-AUC and F1-Score, providing a comprehensive understanding of their predictive power.

Empirical results and insights: The experimental results reveal that XGBoost delivered the strongest overall balance in ROC-AUC, and F1-score, followed by implementation of risk segmentation phase from prediction probability of Logistic Regression. Furthermore, variable-importance analysis identifies **critical predictors** such as loan amount, debt-to-income ratio (DTI), annual income, verification status, and revolving utilisation, offering actionable insights for investors and risk managers.

1.5. Paper outline

Apart from the **Introduction**, this paper is organised as follows. **Section 2** defines the project scope, outlining the aims, objectives, deliverables, stakeholder analysis, and success criteria. **Section 3** presents the methodology, detailing the data preparation pipeline and model-selection process. **Section 4** describes the experimental design, introduces baseline models for comparison, and discusses the results through statistical and visual analysis. **Section 5** concludes the study with a discussion of findings, limitations, and future research directions aimed at strengthening the predictive and practical value of machine learning in the alternative lending industry.

2. Project Scope

The scope of this project is to design and implement a predictive modeling framework for LendingClub for default loans detection using historical data. The project aims to minimize credit risk by developing a binary classification model that distinguishes between default and non-default loans. To achieve the aims, we define clear and specific objectives, deliverables, related stakeholders, and success criteria. The project scope covers the entire project lifecycle from initial proposal, data exploration, to model development, evaluation, and interpretation, ensuring the robustness in technical level and the relevance in business level.

2.1. Project aims

The overarching aim of this project is to develop a practical, data-driven **Loan Default Prediction Model** that enables more accurate risk assessment of LendingClub's loan applicants. By predicting the likelihood of borrower default, the project seeks to strengthen LendingClub's credit evaluation framework, reduce non-performing loans, and enhance profitability while maintaining fair and responsible lending practices.

Ultimately, this project aligns with LendingClub's long-term strategic goals of leveraging artificial intelligence and analytics to **enhance risk management efficiency**, minimize default rates, and sustain investor confidence. The predictive model can be integrated into LendingClub's existing loan approval pipeline, automating parts of the risk evaluation process and allowing analysts to focus on higher-value judgment tasks.

2.2. Project objectives and deliverables

To achieve the stated aim, three key **objectives** have been formulated, each addressing a distinct phase of model development and evaluation:

- **Compare multiple machine learning algorithms** to determine the most effective model for predicting loan defaults. This includes testing and tuning models such as Logistic Regression, LightGBM and XGBoost to identify which provides the best trade-off between interpretability and performance.

- **Evaluate model performance** using comprehensive metrics, including F1-score, AUC-ROC, Precision, Recall, and Confusion Matrix. These metrics are chosen to account for class imbalance in the dataset, ensuring that the model not only predicts accurately overall but also identifies high-risk borrowers effectively.
- **Interpret the model results** to provide workable insights for LendingClub on business level, emphasize how the models could support the improvement of risk management and profitability capability

There are 2 main deliverables for this project, which cover the whole project from preliminary proposal to the final research paper.

- **A proposal** outlining the business context, project scope, initial methodology, project timeline, and group allocations.
- **A comprehensive research paper** covering the project introduction, project scope, methodology, and experiment results.

2.3. Stakeholder analysis

Successful delivery of this project depends on clear communication and alignment among multiple stakeholders-both internal and external to LendingClub. Each stakeholder group plays a distinct role in ensuring that the predictive model is accurate, ethical, and aligned with regulatory expectations. **Seven key stakeholders** have been identified, categorized by their **level of interest** and **influence**.

Senior management team of Lending Club: who are the critical internal stakeholders with high interest and high influence. They are the project initiators, resources allocators, and key decision makers in the project.

- **Risk control team:** as the rule developer and risk control supervisor, they are the critical internal stakeholders with high interest and high influence.

Data Analysis team: as the model developers and technical implementers, they are the internal critical stakeholders with both high interest and high influence.

- **Borrowers/Loan applicants:** as one of the ultimate beneficiary of the project, they are the significant external stakeholders with comparatively low interest during the project development but high influence on the project's success.
- **Customer service team:** they are the significant internal stakeholders with comparatively high interest and low influence fall into, as they need to know the project progress and understand the models to provide efficient and effective service as the frontline communicators.
- **Compliance and legal team:** as the gatekeeper of regulatory compliance and legal risks, they are the minor internal stakeholder with comparatively low interest and influence.
- **Independent regulatory institutions:** as the external supervisors to ensure bank compliance and customers' benefits, they are the minor stakeholders with comparatively low interest and influence.

Stakeholders	Stake	Motivators	Contribution	Information Needs	Communication
Senior Management Team ***	project ROI and the achievement of strategic target	business value ROI of the project increase overall bank profits bank reputation	develop strategy direction provide related resources	project ROI project progress the evaluation of project impact	periodic reports strategic briefings
Risk Control Team ***	loan risk control accuracy bad loans reduction	decrease loan default rate improve the effectiveness of risk control strategies	develop and define the business rules for classifying the loan as defaulted validate the model effect	develop and define the business rules for classify the accuracy of the developed models the internal logic of the loan default evaluation of the loan as defaulted validate the model effect	deep involving in the model developing process regularly review the model performance
Data Analysis Team ***	model performance data quality	technical implementation model optimization	develop the model process the related dataset	the dataset structure the model metrics	high-frequency meetings related technical files
Customer Service Team	customer satisfaction customer service efficiency	improve customer satisfaction improve customer experience	collect and provide customers' feedback execute the improving strategy optimize service process	customer feedback specific service improvement suggestions explanation of the project modelling result	specific case study customer service and project modelling rules related training customer FAQ file
Compliance & Legal Team	regulatory compliance customer privacy protection	comply with related laws and regulations avoid legal risks	compliance with related law	the range of model application the data source	compliance review meetings legal opinions and advisory papers
Borrowers	the loan application result loan status classification the fairness of assessment rules	obtain the loan from bank ensure the loan status is reasonable all the process is fair and transparent	provide related data to the project research provide related feedback	protection of customer rights and interests privacy policy loan application evaluation rules	customer notification customer service agreement
Independent Regulatory Bodies	supervise the regulatory compliance in the whole project the adherence to related industrial standards	compliance check customer rights and benefits protection ensure the stability of financial system	provide regulatory framework review the project and applied models	the regulatory compliance related explanations/clarification of applied models	compliance report compliance communication

Figure 2: Stakeholder Analysis Table

2.4. Success criteria

To better assess whether our project has been completed well, here we have set total 4 key criteria.

- **Improvement in default loans detection:** the developed models could identify the high-risk customers in the early stage to effectively improve defaults loans detection compared to baseline.
- **Improvement in loan approval efficiency:** the risk evaluation could be done automatically, and the manual review could be reduced.
- **Improvement in output interpretability and compliance:** the developed models provide feature importance and interpretable outputs to effectively support the company's regulatory compliance and transparency.
- **Improvement in model accuracy:** the model accuracy of loan default prediction could be improved, with AUC larger than 0.7, and F1-score larger than 0.6.

3. Methodology

3.1. Methodology framework

The methodology adopted in this study is the **CRISP-DM (Cross Industry Standard Process for Data Mining)** framework, which provides a structured and iterative process for developing data-driven models. CRISP-DM consists of six interconnected phases - **Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment** that collectively guide the full data science lifecycle from defining the problem to delivering actionable insights.

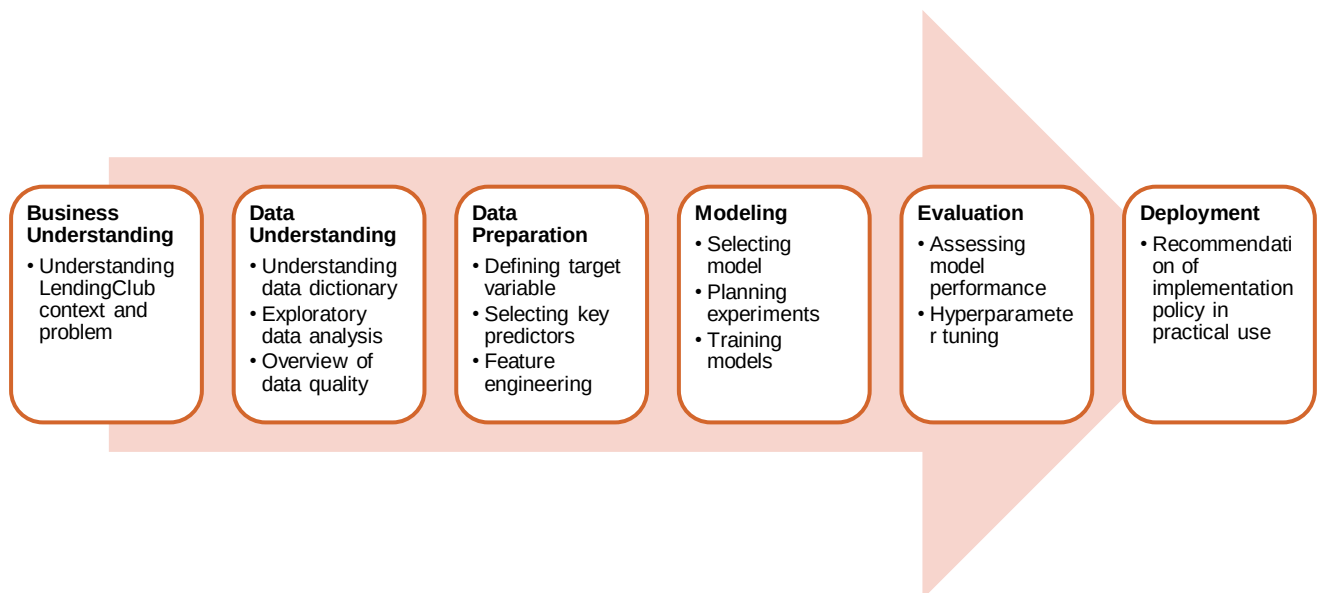


Figure 3: CRISP-DM Methodology

3.2. Data understanding

3.2.1. Dataset overview

The dataset used in this study originates from **LendingClub**, a leading U.S.-based peer-to-peer lending platform that connects borrowers and investors through its online marketplace ([LendingClub Official Website](#)). The data, obtained from **Kaggle's LendingClub Loan Dataset** ([source](#)), contains publicly available loan records issued between **2016 and 2018**, representing one of the most comprehensive open datasets for consumer credit risk analysis. It comprises approximately **1.37 million loan entries** with

151 input variables and one target variable, *loan_default*. This period was chosen to ensure label completeness and maturity for model evaluation.

3.2.2. Target variable definition

In developing the LendingClub Loan Default Prediction Model, we first **standardised the target variable** to ensure it accurately represents credit risk and can be modelled in a **binary classification** setting. The raw *loan_status* field contains multiple granular categories such as “Current,” “Fully Paid,” “Charged Off,” “Late (31–120 days),” and others.

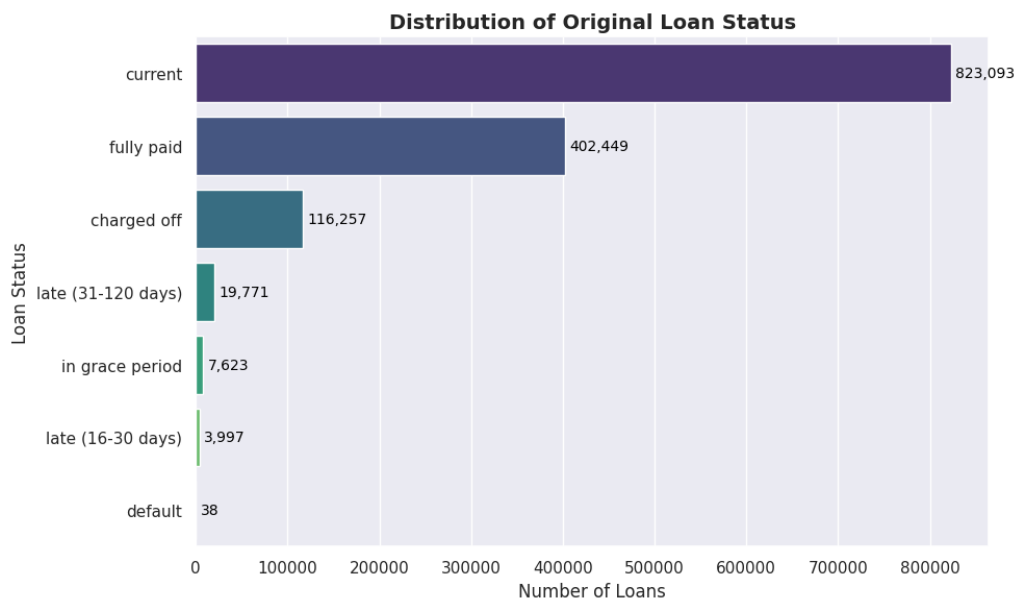


Figure 4: Distribution of original loan status variable

Many of these statuses are either transient, for example “In Grace Period,” or operational rather than final, for example “Current.” Including them would introduce **noise** because these loans have not yet reached an outcome and therefore cannot be labelled reliably.

To address this, we restricted the dataset to loans with a resolved outcome. We defined “non-default” as loans that were fully repaid, and “default” as loans that were charged off or explicitly marked as default. We then mapped these into a **binary loan_default_flag**, where 0 represents non-default and 1 represents default. All loans with ambiguous or ongoing statuses were excluded. This filtering reduced the dataset from 1,373,228 total

observations to 518,744 usable observations, removing 854,484 unresolved loans. The final class distribution is 77.58% non-default and 22.42% default.

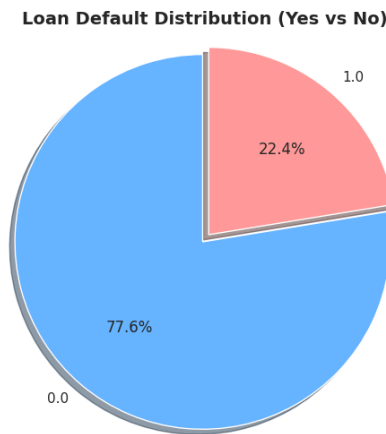


Figure 5: Binary loan default distribution

This approach aligns the prediction task with the real business question: can we forecast which new loans are at risk of ending in default, given known historical outcomes.

3.2.3. Data exploration

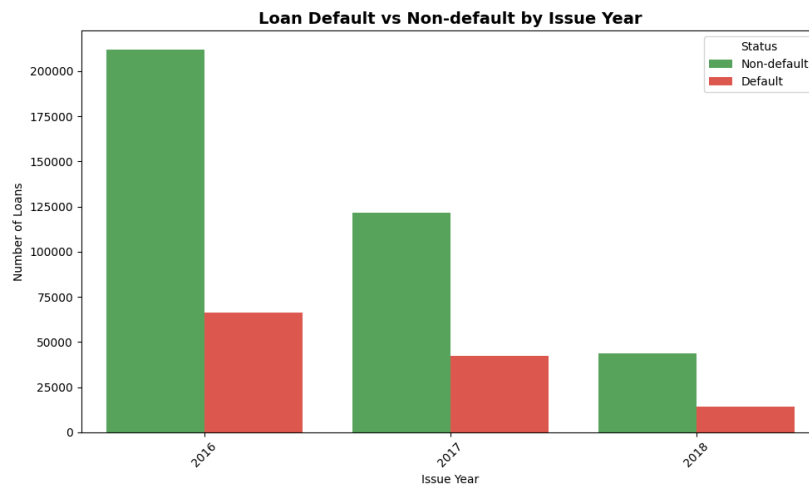


Figure 6: Loan default status distribution by year

Figure 6 shows that both loan issuance and defaults declined from 2016 to 2018. The drop likely reflects smaller loan volumes in later periods rather than a structural change in credit risk. Because the dataset only covers loans issued up to mid-2018, many later loans

have not yet matured (loan defaults depend on maturity, especially longer-term (60-month) loans), so their default counts remain incomplete.

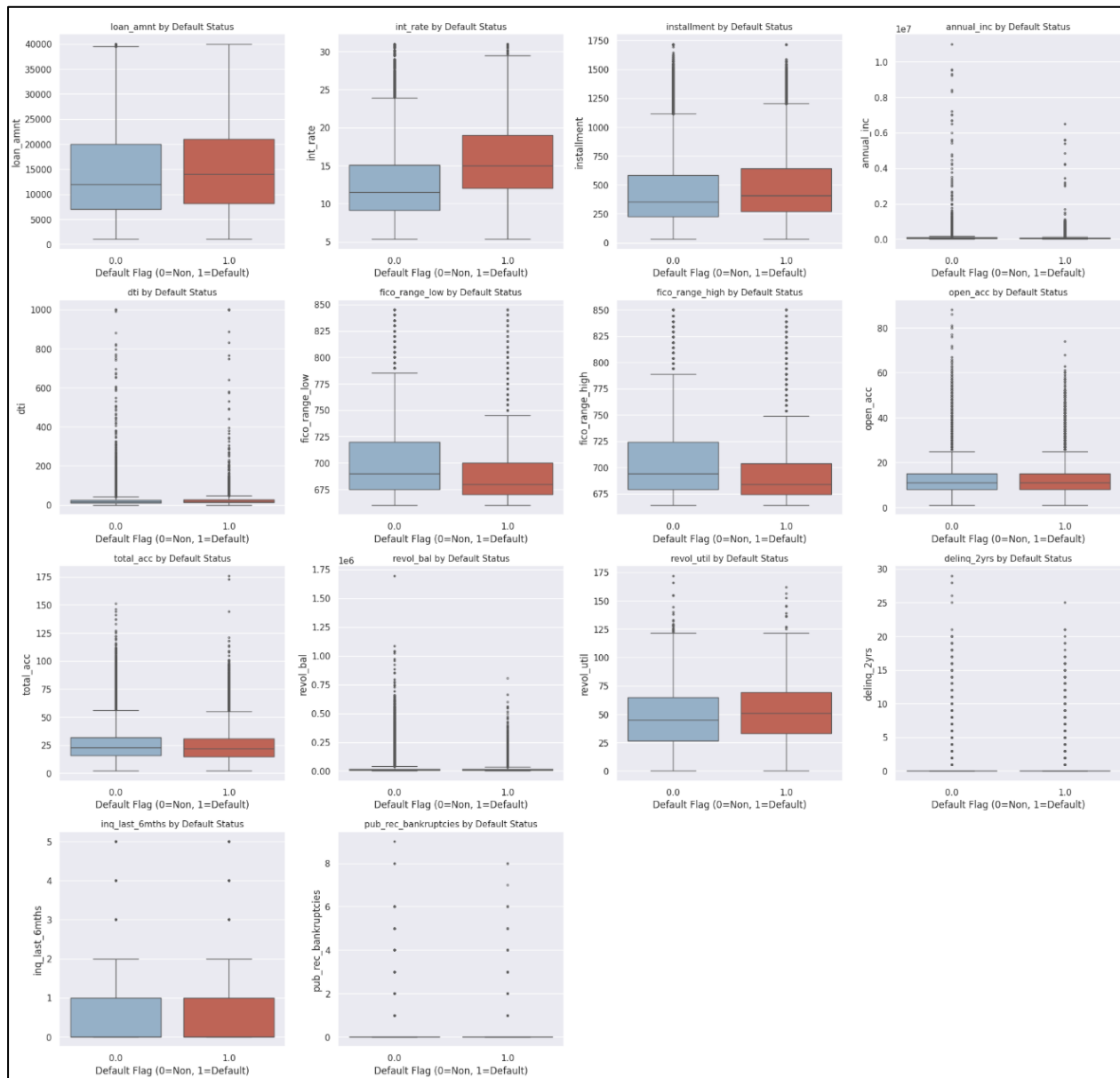


Figure 7: Summary of EDA

The univariate analysis reveals several structural and behavioural risk patterns across LendingClub borrowers. Loan-related variables, including loan amount, interest rate, and installment, display a clear upward shift among defaulted borrowers. This indicates that borrowers taking larger or higher-cost loans are more exposed to repayment difficulty, particularly when interest obligations amplify monthly cash flow stress. These results are consistent with credit-risk

theory, where over-leveraging and adverse selection increase the probability of default among high-rate borrowers (Stiglitz, 1981).

From the affordability perspective, defaulted borrowers show lower annual income and higher debt-to-income ratios, suggesting that affordability constraints and income volatility are key determinants of repayment outcomes. Borrowers with weaker income buffers are more sensitive to economic shocks or employment instability, raising the likelihood of delinquency once obligations exceed manageable thresholds (Richard Disney, 2013).

Credit history indicators offer further validation. Non-defaulted borrowers exhibit higher FICO scores and maintain a greater number of open accounts, reflecting long-standing credit experience, responsible utilisation, and stronger access to credit markets. Conversely, defaulted borrowers tend to exhibit higher revolving credit utilisation, implying tighter liquidity and heavier reliance on short-term borrowing. Though events such as prior delinquencies and bankruptcies are relatively rare, their presence strongly signals long-term credit deterioration.

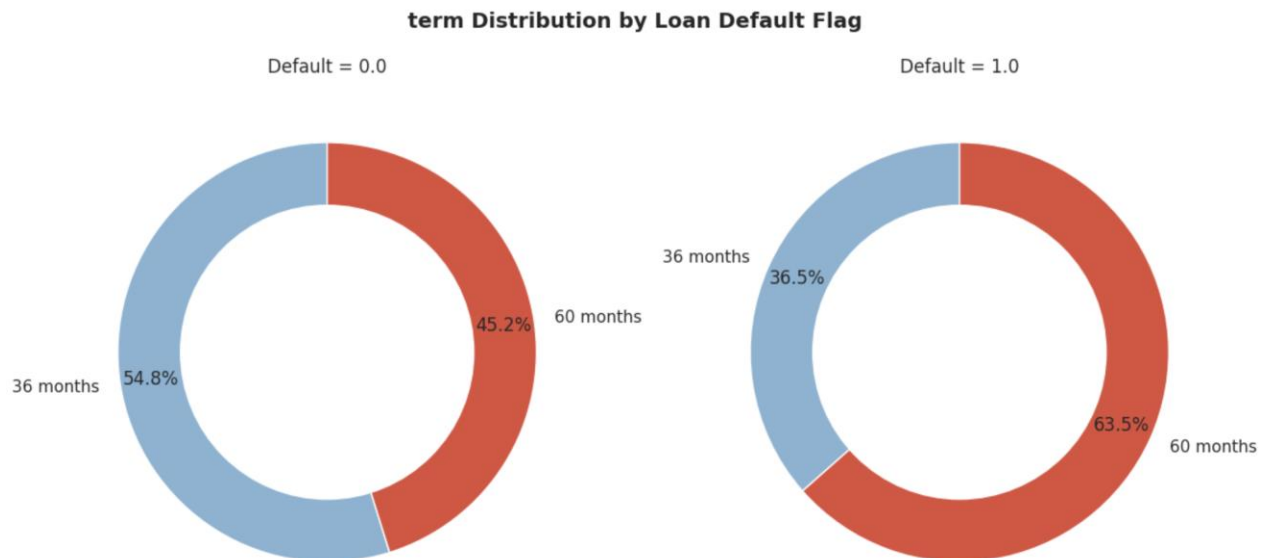


Figure 8: Distribution of Loan Terms (36 vs 60 months) by Default Status

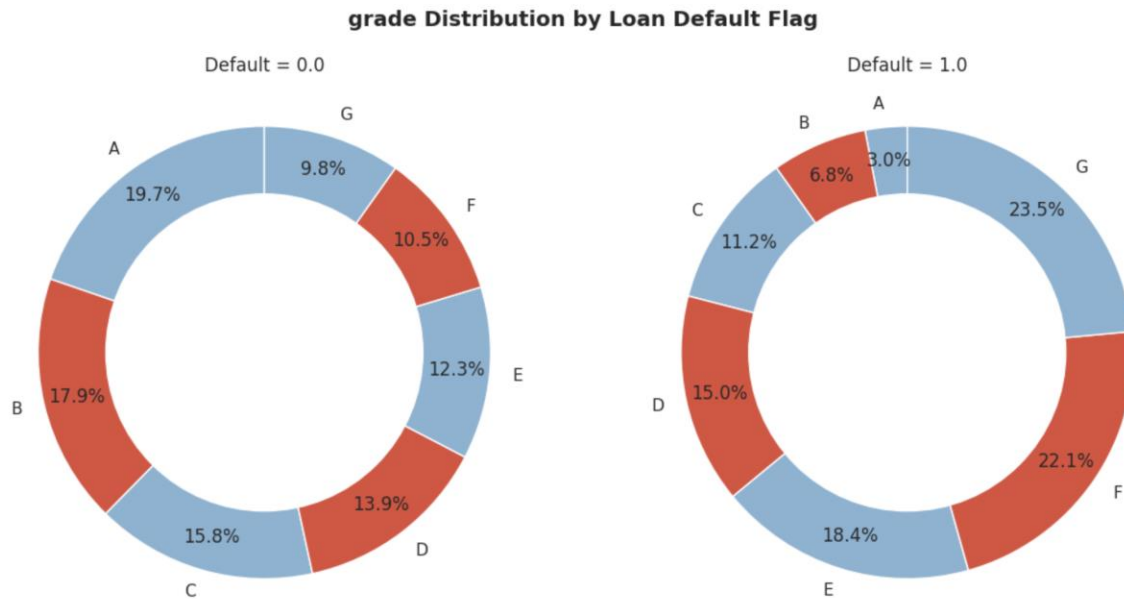


Figure 9: Distribution of grade by Default Status

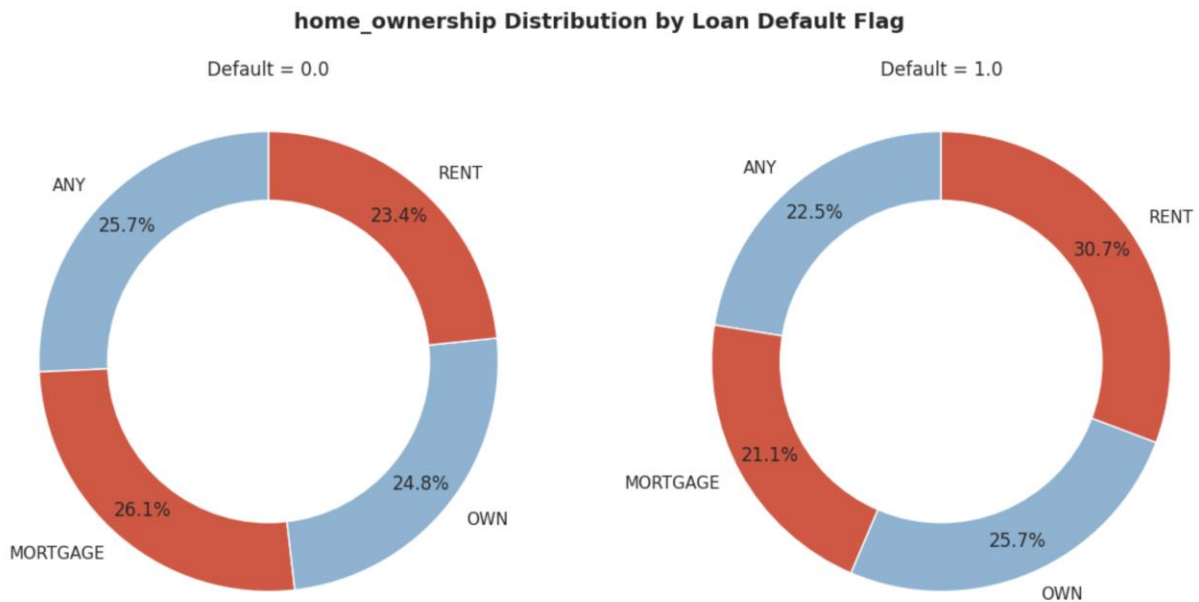


Figure 10: Distribution of home ownership by Default Status

Overall, the categorical analysis highlights three major drivers of default behaviour. Default risk escalates with weaker credit grades (E-G), establishing grade as the clearest signal of borrower quality. Longer loan terms (60 months) also correspond to greater default likelihood, reflecting higher exposure and repayment burden. Meanwhile, renters default more than homeowners or mortgage holders, signalling weaker financial stability.

3.3. Modelling approach

3.3.1. Rationale for model selection

We evaluate three model families that align with the structure of the data and the regulatory expectations of credit decisioning: Logistic Regression, XGBoost and LightGBM.

Study	Focus Area	Methods/Models	Key Findings
[1]	Mobile-based lending	Decision Trees, SVM, Logistic Regression	Decision trees performed well in handling categorical variables and non-linear relationships.
[3]	Model comparison	Gradient Boosting, Random Forest, Logistic Regression	Gradient Boosting and Random Forest achieved significantly higher accuracy and F1-scores.
[8]	Ensemble learning	Voting classifier (Gradient Boosting, Random Forest, XGBoost)	Ensemble methods yielded superior results compared to individual models, enhancing robustness.
[7]	Class imbalance and interpretability	SMOTE, Ensemble Learning, XAI	SMOTE combined with ensemble learning improved recall for the minority class (defaulters). Post hoc XAI methods are needed for transparency.
[10]	Interpretability vs. Performance	Insolvency prediction models	Highlighted the trade-off between model performance and interpretability, which is crucial for regulatory compliance.

Figure 11: Comparative Summary of Key Literature Findings on Loan Default Prediction (Xinyu Zhang, 2025)

Logistic Regression is included because it remains the **standard approach** in retail credit scoring (Sifrain, 2023). It models the log-odds of default as a function of borrower capacity, credit history, and loan contract terms, and its **coefficients** can be directly interpreted in policy and compliance reviews, which is critical when justifying lending decisions to risk governance.

XGBoost is selected because default risk is rarely linear. The interaction between factors such as revolving utilisation, delinquency history, and debt-to-income ratio is often non-additive, and boosted trees can learn these **nonlinear credit risk patterns** and rank-order borrowers by likelihood of default with high discrimination (Xinyu Zhang, 2025).

LightGBM is included to address portfolio scale. Our dataset is large, heterogeneous, and imbalanced; LightGBM's **efficient histogram-based leaf-wise growth** improves both computational efficiency and AUC in high-volume lending environments while preserving the ability to model complex borrower behavior (Ke, 2017). Additionally, LightGBM provides exceptional speed and resource efficiency, which is critical when running the project in a cloud environment such as Google Colab.

3.3.2. Feature selection

The predictor set was restricted to economically meaningful variables that would be available at origination time and would not leak post outcome information.

```
cols_to_keep = [

    # ===== Loan Characteristics =====
    'issue_d', 'issue_year',
    'loan_default_flag',
    'loan_amnt',          # Loan size – larger loans = higher repayment burden
    'term',              # Loan duration – longer = higher risk
    'int_rate',          # Interest rate reflects borrower creditworthiness
    'installment',       # Monthly installment amount
    'grade',             # Lender-assigned risk categories
    'purpose',           # Reason for borrowing (debt consolidation, car, etc.)
    'initial_list_status', # Institutional vs fractional funding

    # ===== Borrower Profile / Socioeconomic =====
    'annual_inc',        # Borrower's annual income
    'home_ownership',    # Own / Mortgage / Rent
    'verification_status', # Whether income was verified

    # ===== Credit History and Behavior =====
    'dti',               # Debt-to-income ratio
    'fico_range_low', 'fico_range_high', # Credit score range (lower/higher)
    'open_acc',          # Number of open credit lines
    'total_acc',         # Total credit accounts
    'revol_bal',         # Revolving balance outstanding
    'revol_util',        # Revolving utilization rate
    'delinq_2yrs',       # Number of delinquencies in past 2 years
    'inq_last_6mths',    # Number of recent credit inquiries
    'pub_rec',           # Public derogatory records (judgments, etc.)
    'pub_rec_bankruptcies', # Bankruptcies count
    'collections_12_mths_ex_med', # Collection events excluding medical
    'chargeoff_within_12_mths', # Charge-offs within past year

]

print(f"Total selected features: {len(cols_to_keep)}")
print(cols_to_keep)

Total selected features: 26
```

Figure 12: Feature selection

Loan characteristics capture contract structure and pricing of risk: *loan_amnt*, *term*, *int_rate*, *installment*, *grade*, *purpose*, *initial_list_status*, and *issue_year* and *issue_d* for vintage effects. These reflect repayment burden, lender assessed credit quality, intended use of funds, and channel (Edward I Altman, 1998).

Borrower profile variables describe capacity to repay: *annual_inc*, *home_ownership*, and *verification_status*, which together proxy income stability and collateral access.

Credit history and behaviour variables quantify existing leverage and recent stress: *dti*, *fico_range_low*, *fico_range_high*, *open_acc*, *total_acc*, *revol_bal*, *revol_util*, *delinq_2yrs*, *inq_last_6mths*, *pub_rec*, *pub_rec_bankruptcies*, *collections_12_mths_ex_med*, and *chargeoff_within_12_mths*.

These features characterise credit utilisation, delinquency patterns, inquiries, public derogatory records, and previous charge offs. The selected inputs align with lender underwriting logic, preserve interpretability, and allow a direct link between model output and actionable credit policy (D. J. HAND, 1997).

3.3.3. Data preparation

3.3.3.1. Data splitting

Loan Maturity Dependency

LendingClub data are inherently time-dependent, and loans issued closer to the dataset's cutoff date - particularly in 2018 - have not existed long enough for default events to fully materialize. This problem is called right-censoring, in which different approaches need to be applied.

Time-Based Data Splitting

To prevent temporal leakage and ensure label completeness, the dataset was partitioned **chronologically** by issue year.

```
# Split by year
train_df = df[df["issue_year"] == 2016].copy()
val_df   = df[df["issue_year"] == 2017].copy()
test_df  = df[df["issue_year"] == 2018].copy() # optional

print("Train:", train_df.shape)
print("Validation:", val_df.shape)
print("Test:", test_df.shape)
```



```
Train: (292878, 26)
Validation: (168988, 26)
Test: (56093, 26)
```

Figure 13: Data splitting

Loans issued in **2016** were used for model training, as this subset contained the largest and most mature loan population, allowing sufficient time for defaults to manifest. The

2017 loans were allocated to validation for parameter tuning and model selection, while the **2018** subset served as a held-out test set to assess out-of-time performance. This temporal split replicates **real-world deployment conditions**, where credit models are trained on historical data and applied to future loan vintages. By training on the most complete vintage (2016), the model benefits from balanced and fully realized default outcomes, mitigating right-censoring bias and improving the reliability of performance evaluation.

3.3.3.2. Data cleaning

During data quality review, we assessed completeness, duplication, and extreme values before modelling.

Missing values

A **complete-case approach** was applied to the modelling dataset. After standardising the target label and restricting the data to 26 features observable at application time, 518,744 records were retained. A strict row-wise missing value filter using `dropna()` removed 785 records, resulting in a final sample of 517,959. The excluded observations accounted for only 0.15% of the data, a negligible proportion that did not affect class balance or temporal consistency.

Duplicates

A duplicate check returned **zero duplicated records**, which indicates that each loan represents a unique lending decision.

Outliers

Outliers were examined within the selected modelling features in the training set using the interquartile range (IQR) to identify extremes and measure their proportion per variable. Each variable was classified into Low, Medium, or High policy groups. Although tree-based models such as LightGBM and Random Forest are inherently robust to outliers, this assessment ensured comparability across all models, particularly Logistic Regression, which is more sensitive to extreme values. Results showed minimal impact of outliers on model performance.

	Feature	Pre Outlier %	Post Outlier %	Δ Outlier % (pp)	Outlier Group	Policy	Q1	Q3	Lower Bound	Upper Bound	Outlier Count
11	delinq_2yrs	20.747205	20.747205	0.000000	High	Keep (likely signal, not noise)	0.00	0.00	0.000	0.000	60764
13	pub_rec	20.015843	20.015843	0.000000	High	Keep (likely signal, not noise)	0.00	0.00	0.000	0.000	58622
14	pub_rec_bankruptcies	13.971346	13.971346	0.000000	High	Keep (likely signal, not noise)	0.00	0.00	0.000	0.000	40919
9	revol_bal	6.632796	0.000000	6.632796	Medium	Log1p if positive else winsorize	5759.00	19283.00	-14527.000	39569.000	19426
3	annual_inc	4.822144	1.914449	2.907695	Medium	Log1p if positive else winsorize	48000.00	94000.00	-21000.000	163000.000	14123
12	inq_last_6mths	4.405247	0.000000	4.405247	Medium	Log1p if positive else winsorize	0.00	1.00	-1.500	2.500	12902
6	fico_range_high	3.335860	0.000000	3.335860	Medium	Log1p if positive else winsorize	674.00	714.00	614.000	774.000	9770
5	fico_range_low	3.335860	0.000000	3.335860	Medium	Log1p if positive else winsorize	670.00	710.00	610.000	770.000	9770
2	installment	2.860577	1.394096	1.466481	Medium	Log1p if positive else winsorize	244.69	599.96	-288.215	1132.865	8378
7	open_acc	2.724343	0.604006	2.120337	Medium	Log1p if positive else winsorize	8.00	15.00	-2.500	25.500	7979
1	int_rate	2.696686	0.000000	2.696686	Medium	Log1p if positive else winsorize	9.17	15.59	-0.460	25.220	7898
15	collections_12_mths_ex_med	1.972494	0.000000	1.972494	Low	Winsorize at IQR bounds	0.00	0.00	0.000	0.000	5777
8	total_acc	1.804847	0.000000	1.804847	Low	Winsorize at IQR bounds	16.00	32.00	-8.000	56.000	5286
16	chargeoff_within_12_mths	0.831404	0.000000	0.831404	Low	Winsorize at IQR bounds	0.00	0.00	0.000	0.000	2435
0	loan_amnt	0.660002	0.000000	0.660002	Low	Winsorize at IQR bounds	7500.00	20000.00	-11250.000	38750.000	1933

Figure 14: Outlier categorization and handling policy

For **Low-group** variables (e.g., *loan_amnt*), values were winsorised to IQR bounds to maintain economic realism. **Medium-group** variables, typically monetary or behavioural counts, were capped at the 99th percentile or LendingClub limits (e.g., 60,000) and log-transformed where appropriate. **High-group** variables, such as prior delinquencies and public records, were retained but capped (e.g., *delinq_2yrs* ≤ 5). This approach suppressed unrealistic extremes while preserving genuine risk signals, improving stability and interpretability in downstream modelling.

3.3.3.3. Class imbalance analysis

The LendingClub dataset shows a strong class imbalance, with non-default loans far exceeding defaults. To mitigate bias toward the majority class, class-weighting strategies were applied across models (Yujia Chen, 2024). **Logistic Regression** used `class_weight="balanced"` to adjust weights inversely to class frequencies, while **XGBoost** and **LightGBM** applied `scale_pos_weight`, calculated as the ratio of negative to positive samples in the training set. This penalised misclassified defaults and improved learning in the minority class. **LightGBM** also employed early stopping to prevent overfitting. These methods enhanced recall on default cases while maintaining precision, achieving more stable and equitable performance. Furthermore, we also evaluate performance using **class sensitive metrics** including recall, precision, F1 score and AUC (GeeksforGeeks, 2025).

3.3.3.4. Feature encoding

To prepare the data for modelling, we constructed a supervised preprocessing pipeline that treats numeric and categorical predictors appropriately while preventing information leakage.

```
Ordinal features: ['term']
One-hot features: ['home_ownership', 'purpose', 'verification_status', 'initial_list_status', 'grade']
Numeric features: 17 columns
Linear shapes: (292878, 47) (168988, 47) (56093, 47)
Tree shapes: (292878, 47) (168988, 47) (56093, 47)
```

Figure 15: Feature groups

First, features were grouped into numeric variables, ordinal categorical variables, and nominal categorical variables. The loan term variable is ordinal, so we applied an OrdinalEncoder with an explicit order, 36 months then 60 months, which reflects increasing repayment horizon and therefore increasing repayment stress. Other categorical variables such as home_ownership, purpose, verification_status, initial_list_status, and grade were one hot encoded after consolidating rare levels into an “Other” category. This reduces the dimensionality explosion from infrequent categories and improves model stability at inference time. Numeric features were either standardised using StandardScaler for linear models, or passed through unchanged for tree-based models, since tree learners do not require scaling.

```
num_cols = 17
ordinal_feats = ['term']
onehot_feats = ['home_ownership', 'purpose', 'verification_status', 'initial_list_status', 'grade']
OHE categories per col: [4, 13, 3, 2, 7] | total = 29
Expected total dims = 47
```

Figure 16: Results after encoding

The final transformed training matrices contained 47 engineered columns for the linear pipeline and 29 interpretable plus numeric inputs for the tree-based pipeline, confirming reproducible, model specific feature representations.

3.3.3.5. Data Partitioning

After final cleaning, the dataset was partitioned into training, validation, and test sets to support model development, hyperparameter tuning, and unbiased performance assessment. For each split, the binary target variable *loan_default_flag* was isolated as *y*, and the remaining predictors were stored as *X*. The same feature set was used across all splits to ensure consistency at inference time.

We then profiled the predictors and identified two data types for downstream preprocessing: **17 numeric features**, such as income, FICO range, and utilisation, and **6 categorical features**, such as home ownership and loan grade. This explicit separation enables model-specific encoding, scaling, and evaluation under a controlled, reproducible pipeline.

4. Experiments

4.1. Experiment plan

4.1.1. Experiment objectives

This experiment aims to evaluate and compare multiple machine learning models for predicting **loan default risk** in the **LendingClub** dataset. The goal is to identify the most effective model that balances **accuracy**, **recall**, and **computational efficiency**, supporting risk mitigation strategies for financial institutions. The dataset was split **chronologically** into training, validation, and testing sets to avoid temporal leakage and ensure realistic evaluation. No synthetic oversampling was applied, maintaining model robustness under real-world imbalance. All experiments were conducted in **Python**, using **scikit-learn**, **XGBoost**, and **LightGBM** frameworks.

4.1.2. Experiment phases

This experiment consists of 4 main phases.

4.1.2.1. Baseline model establishment

The Logistics Regression is applied as the benchmark to establish **baseline performance**. This allows for a clear assessment of the added values of more complex models in subsequent phases (Ellis, 2021).

4.1.2.2. Model training and evaluation

After the baseline establishment was completed, train total 3 applied models - Logistic Regression, XGBoost, and LightGBM on the splitting training dataset, evaluated on the testing dataset by selected metrics such as accuracy, precision, recall, F1-score, and AUC. However, due to the dataset's imbalance and the model's complexity, recall, AUC, and F1 will be focused more on evaluation.

4.1.2.3. Model comparison

According to the model training and evaluation results completed in step 2, the performance of all 3 selected models is compared to identify the one with best performance. This step highlights the tradeoffs between model complexity and prediction ability, helps to select the most suitable model for our default loans prediction.

4.1.3. Model implementation plan

Total three models are applied in our project, including Logistics Regression, XGBoost, and LightGBM. Related details shown in below points.

4.1.3.1. Logistic Regression implementation

Logistic Regression is selected as the baseline linear model due to its simplicity and interpretability. To address the class imbalance in default loans prediction, `class_weight='balanced'` is applied to adjust the weights inversely proportional to class frequencies, ensuring the minority class which is the default loans is not neglected.

The solver is set to 'saga' because it could handle large and sparse datasets efficiently, which is necessary for high-dimension data with many categorical features. Here we use `penalty="l2"` to avoid overfitting and ensure model stability

```
# === 1 Train Logistic Regression (SAGA solver) ===
lr = LogisticRegression(
    solver="saga",
    penalty="l2",
    C=1,
    class_weight="balanced",
    max_iter=2000,
    tol=1e-3,
    n_jobs=-1,
    random_state=42
)
lr.fit(x_tr, y_train)
```

Figure 17: Parameters for training logistic regression

The predictions were generated for validation and testing dataset, and performance was assessed applying standard classification metrics including ROC-AUC, PR-AUC, LogLoss, Accuracy, Precision, Recall, and F1-score. Detailed evaluation results are presented in the subsequent Performance Evaluation Metrics parts.

4.1.3.2 XGBoost implementation and tuning

In our project, XGBoost was selected for its ability to identify non-linear relationships among features. To address the class imbalance, the parameter `scale_pos_weight` was computed based on the ratio of negative to positive samples in the training dataset, by which improving Recall for default loans identification and decreasing the risk of overlooking the high-risk loan borrowers.

```
# =====
# 1 Handle class imbalance
# =====
scale_pos_weight = (len(y_train) - y_train.sum()) / y_train.sum()
print(f"scale_pos_weight = {scale_pos_weight:.2f}")
```

Figure 18: Class imbalance handling in XGBoost

The base model was defined with `objective='binary:logistic'` and `eval_metric='auc'`, which because AUC is robust under imbalanced situations. The `tree_method='hist'` was selected for its efficiency on large dataset.

```

# 2 Define base model + parameter grid
# =====
xgb_model = XGBClassifier(
    objective='binary:logistic',
    eval_metric='auc',
    scale_pos_weight=scale_pos_weight,
    tree_method='hist',
    random_state=42,
    n_estimators=500
)

```

Figure 19: XGBoost parameter for training

The hyperparameter tuning was performed via RandomizedSearchCV over parameters such as max_depth and min_child_weight to control complexity and prevent possible overfitting, subsample and colsample_bytree to introduce randomness for better generalization, learning_rate to balance convergence speed and stability, and gamma to add regularization to reduce overly complex dataset splits.

```

param_grid = {
    'max_depth': [4, 5, 6, 7, 8],
    'min_child_weight': [1, 3, 5, 7, 10],
    'subsample': [0.6, 0.7, 0.8, 0.9],
    'colsample_bytree': [0.6, 0.7, 0.8, 0.9],
    'gamma': [0, 0.1, 0.3, 0.5],
    'learning_rate': [0.01, 0.03, 0.05, 0.1]
}

search = RandomizedSearchCV(
    estimator=xgb_model,
    param_distributions=param_grid,
    n_iter=20,
    scoring='roc_auc',
    n_jobs=-1,
    cv=3,
    verbose=2,
    random_state=42
)

```

Figure 20: Hyperparameter tuning for XGBoost

Early stopping was used during retraining to prevent overfitting, applying validation AUC as the metric. The optimal threshold was determined using the precision-recall curve, and the final model along with metadata was saved for future deployment.

4.1.3.3. LightGBM implementation

LightGBM was selected for its efficiency in handling large datasets and categorical features. To handle class imbalance, scale_pos_weight was computed based on the ratio of negative to positive samples, which was similar to XGBoost.

```

# 1 Compute class imbalance weight
# =====
scale_pos_weight = (len(y_train) - y_train.sum()) / y_train.sum()
print(f"scale_pos_weight = {scale_pos_weight:.2f}")

# Ensure numeric
y_train = np.asarray(y_train).astype(int)
y_val = np.asarray(y_val).astype(int)
y_test = np.asarray(y_test).astype(int)

```

Figure 21: LightGBM's handling class imbalance

The model was initialized with key parameters such as `boosting_type='gbdt'` for strong prediction ability, `learning_rate=0.05` and `n_estimators=500` to tune for convergence and performance, `num_leaves=31` and `max_depth=-1` to balance complexity and avoid overfitting, and `feature_fraction` and `bagging_fraction` for regularization and faster training.

```

# 2 Initialize model
# =====
lgb_clf = LGBMClassifier(
    objective="binary",
    boosting_type="gbdt",
    learning_rate=0.05,
    num_leaves=31,
    max_depth=-1,
    feature_fraction=0.8,
    bagging_fraction=0.8,
    bagging_freq=5,
    lambda_l2=1.0,
    scale_pos_weight=scale_pos_weight,
    random_state=42,
    n_estimators=500
)

```

Figure 22: LightGBM parameters for training

Early stopping was implemented during training to prevent overfitting, using validation AUC as the monitoring metric. The optimal classification threshold was determined on the validation set by maximizing the F1-score using the precision-recall curve. Finally, the tuned model was evaluated on the test set and visualized with ROC and PR curves.

4.1.4. Performance evaluation metrics

The performance of Logistic Regression, XGBoost, and LightGBM was evaluated on both the validation and testing dataset using metrics including ROC-AUC, PR-AUC, Accuracy, Precision, Recall, and F1-score. The result comparison table is shown in the subsequent result part.

Table 1: Evaluation metrics

Metric	Description
ROC-AUC	Measure the model's ability to distinguish between default and non-default loans across thresholds, helping LendingClub and other similar financial institutions to provide loan default risk alerts timely and effectively.
Recall	Proportion of default loans were correctly identified. High Recall indicates less risky loans and the potential losses reduced as a result.
Precision	Proportion of predicted default loans were truly default. High Precision prevents unnecessary inventions or restrictions on high-quality customers and the customer experience could be improved as a result.
PR-AUC	Area under Precision-Recall curve, useful for imbalanced dataset like the default loans in our project, ensuring timely and accurate identification of high-risk loan borrowers.
F1-score	Harmonic mean of precision and recall, balancing timely default loans detection with fewer inaccurate alarms for LendingClub's risk management.
Accuracy	Overall correctness of default loans predictions.

Visualizations such as ROC and Precision–Recall curves are provided in the following Results part for further interpretability.

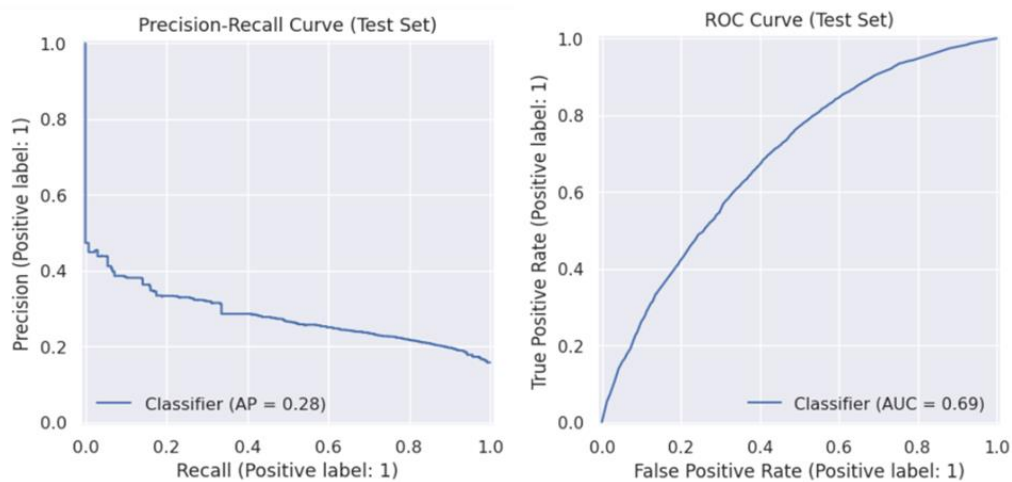


Figure 23: Visualisation of ROC and Precision-Recall curve

4.2. Results

4.2.1. Model performance on validation and test sets

Three models were evaluated: Logistic Regression, tuned XGBoost, and LightGBM.

Table 2: Model Performance Results

Metric	Logistic Regression	LightGBM	Tuned XGBoost
ROC-AUC (Validation)	0.7037	0.694	0.7116
ROC-AUC (Test)	0.701	0.689	0.709
Recall (Validation, Default=1)	0.691	0.6778	0.6983
Recall (Test, Default=1)	0.6708	0.6655	0.6708
Precision (Validation, Default=1)	0.345	0.34	0.35
Precision (Test, Default=1)	0.25	0.24	0.25
PR-AUC (Validation)	0.4008	0.3866	0.4144
PR-AUC (Test)	0.289	0.279	0.305
F1 (Validation)	0.46	0.454	0.465
F1 (Test)	0.363	0.353	0.368
Accuracy (Validation)	0.625	0.62	0.628
Accuracy (Test)	0.629	0.62	0.638

In terms of ROC AUC, tuned XGBoost achieved the strongest performance, with 0.711 on validation and 0.709 on test. A ROC AUC near 0.70 means the model can rank borrowers by default risk: given two applicants, one who will default and one who will not, the model usually **assigns a higher risk score to the future defaulter**. This is fundamental for credit screening and risk-based pricing. Logistic Regression performs similarly, with ROC AUC 0.704 (validation) and 0.701 (test). LightGBM is slightly lower, at 0.694 and 0.689. The difference between validation and test for all models is below 0.02, indicating limited overfitting and good robustness for deployment.

Recall is approximately 0.67, meaning the model can detect around **two thirds** of eventual defaulters before funds are released. Precision is about 0.25, meaning that one in four borrowers flagged as high risk will actually default. This is acceptable if high risk applicants are routed to manual review or tighter pricing, rather than automatically rejected.

On the test set, tuned XGBoost reaches a PR AUC of approximately 0.306. This almost doubles the concentration of true future defaulters in the highest risk segment compared

with a random review strategy. Logistic Regression and LightGBM show slightly lower but comparable PR AUC.

4.2.2. Strengths and weaknesses observed in predictions

All three models achieve recall around 0.66 to 0.70 for the positive class, meaning they correctly identify roughly two thirds of borrowers who ultimately default. This is important for **loss prevention** because missed defaulters directly translate to realised credit loss. However, precision on the default class is around 0.25 on the held-out test set, which means that only one in four applicants flagged as high risk would actually default. This is an expected pattern in imbalanced consumer credit data where true defaults are relatively rare. The F1 scores between 0.35 and 0.37 on test indicate a workable balance between capturing risky borrowers and limiting false alarms.

From a model governance perspective, Logistic Regression is transparent and stable but captures only linear effects. XGBoost achieves the **best tradeoff** between recall and precision, and the highest ROC AUC and F1, but is less directly interpretable without post hoc tools. LightGBM performs similarly to XGBoost in recall but with slightly lower precision and ROC AUC, which suggests that its split strategy is marginally less aligned with the structure of this portfolio.

4.2.4. Interpretation of key drivers

Logistic Regression provides direct interpretability through its coefficients.

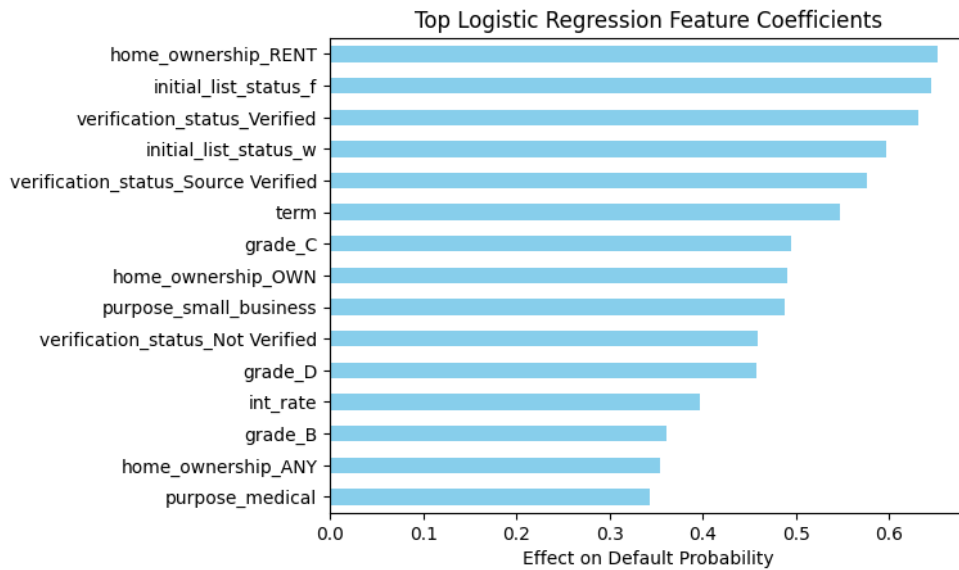


Figure 24: Top Logistic Regression Feature Coefficients

The most **influential predictors** increasing default probability are related to borrower profile, loan channel, and documentation quality. **Renting status** (home_ownership_RENT) is one of the strongest risk indicators, suggesting that applicants without housing assets are more likely to miss repayments. Loans that are listed through particular funding **channels** and loans where income **verification** is weaker show higher predicted default. This means origination quality and documentation quality matter, not only raw credit history. Certain loan **purposes**, such as small business, and lower internal credit grades, such as **grade C**, are also linked to elevated default probability.

▼ Least influential features:

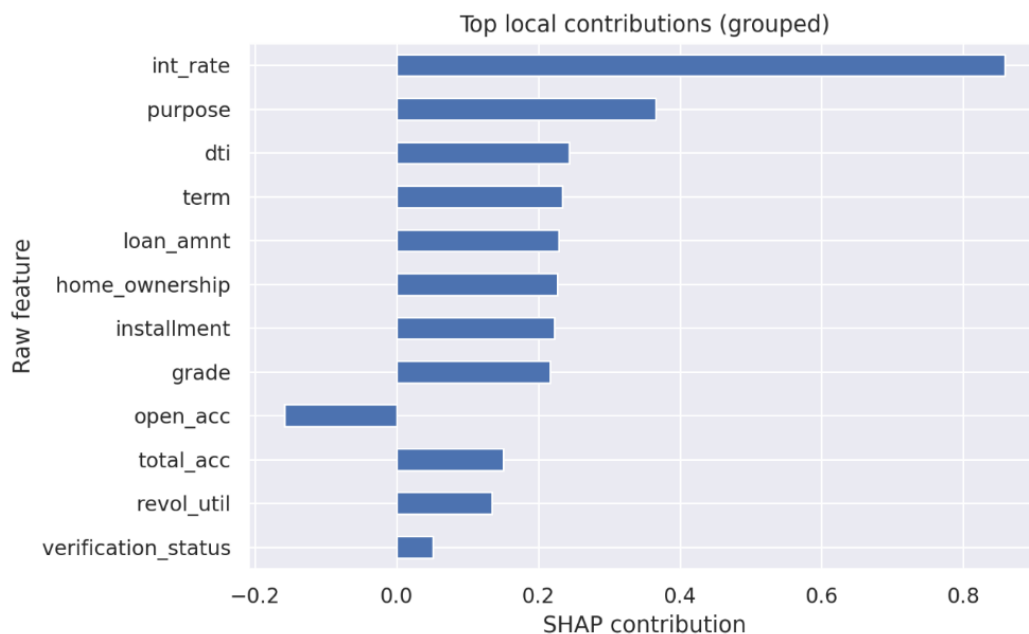
	Feature	Coefficient	Abs_Coefficient
37	grade_A	0.045235	0.045235
38	delinq_2yrs	0.032334	0.032334
39	collections_12_mths_ex_med	0.017387	0.017387
40	pub_rec	0.015498	0.015498
41	purpose_wedding	-0.013819	0.013819
42	revol_util	0.013691	0.013691
43	purpose_house	0.013329	0.013329
44	purpose_car	0.004038	0.004038
45	pub_rec_bankruptcies	-0.000832	0.000832
46	chargeoff_within_12_mths	-0.000364	0.000364

Figure 25: Least influential features

In contrast, **traditional bureau style delinquency indicators** such as public record counts, bankruptcies, and delinq_2yrs appear with comparatively **small absolute coefficients** in the fitted linear model.

The precision recall and ROC curves confirm that the model meaningfully separates high risk from low-risk segments, although with diminishing precision at very high recall.

We also have SHAP analysis for XGBoost:



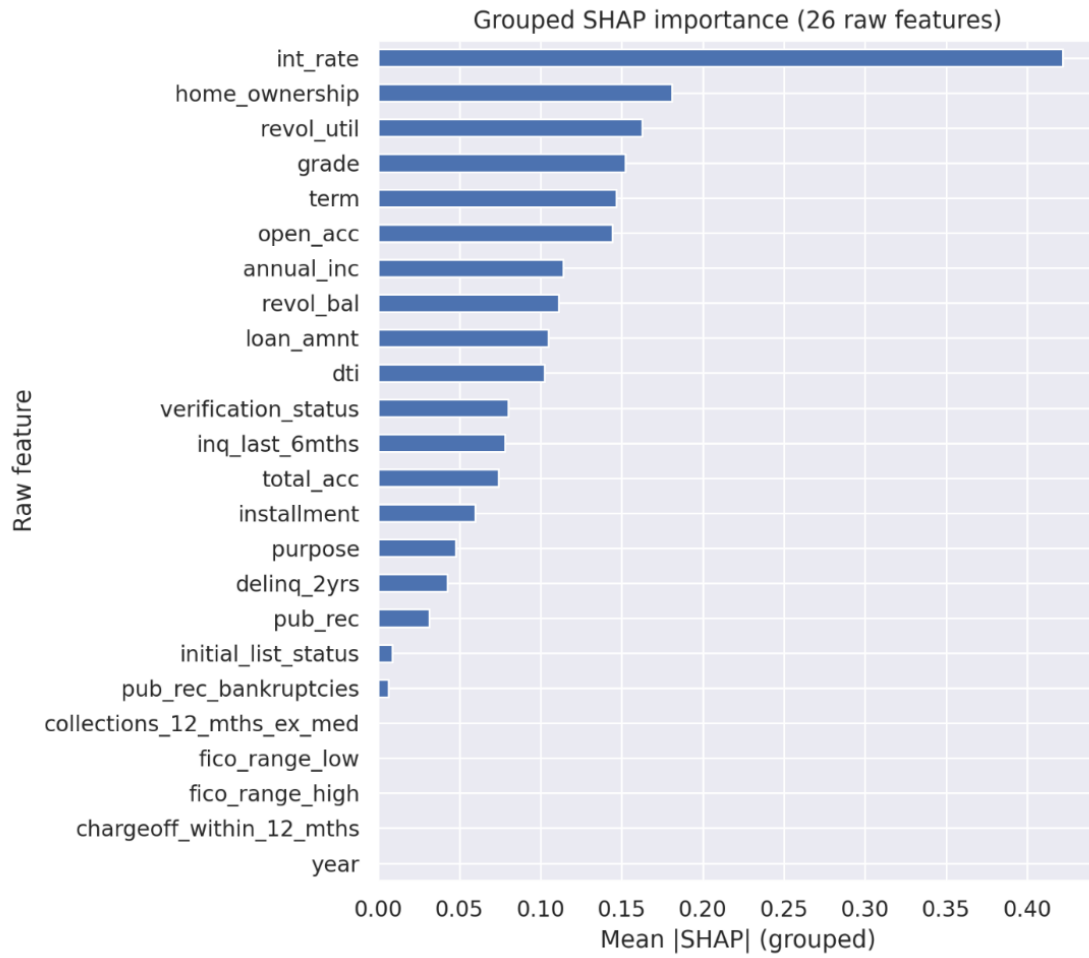


Figure 26: Grouped SHAP Importance

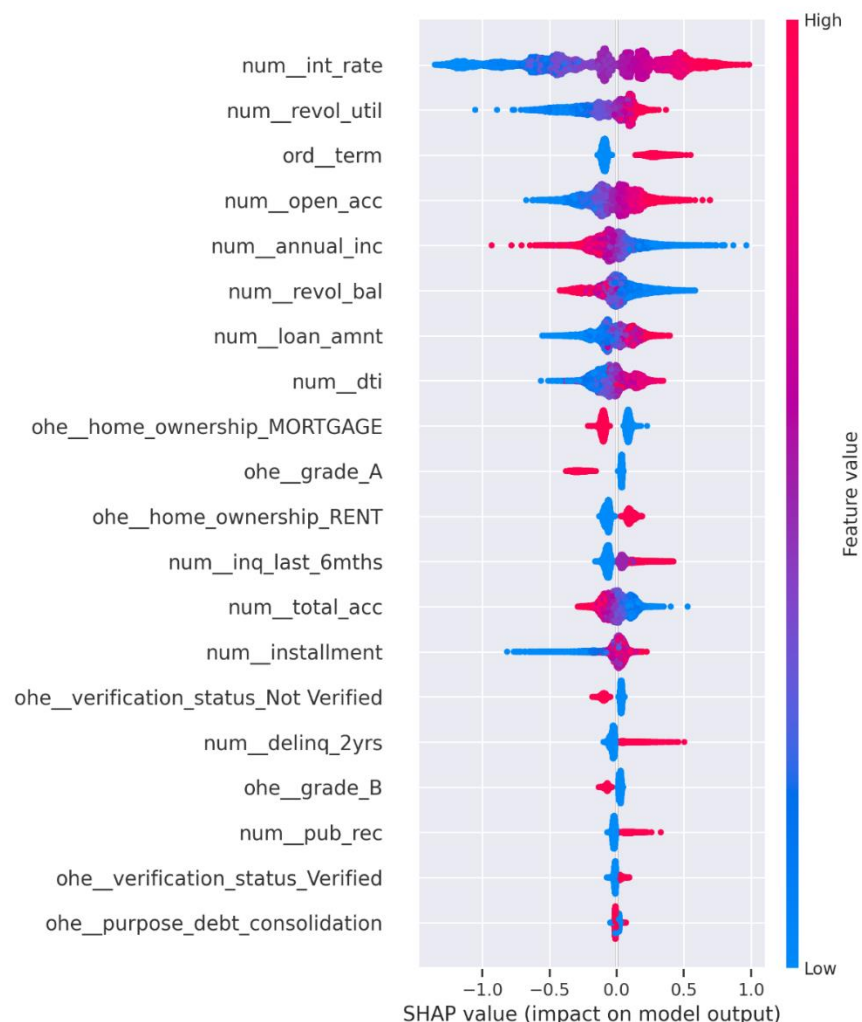


Figure 27: SHAP value

Based on the SHAP analysis of the XGBoost model (grouped by raw features), **int_rate** is the dominant driver of default risk, followed by **home_ownership**, **revol_util**, **grade**, **term**, **open_acc**, **annual_inc**, **revol_bal**, **loan_amnt**, and **dti**. The beeswarm plot shows that **higher values** of **int_rate**, **revol_util**, **loan_amnt**, **term** (60 months), **dti**, and **revol_bal** generally **push SHAP values positive**, increasing risk; conversely, **higher annual_inc** and higher-quality attributes such as **grade A/B**, **home_ownership = MORTGAGE**, and **verification_status = Verified** tend to have negative SHAP values, **reducing risk**. Credit depth variables (**open_acc**, **total_acc**) are broadly protective (negative contributions), whereas **inq_last_6mths** is positive, indicating that many recent inquiries raise risk. In the local (case-level) example, the high-risk score is driven primarily by **int_rate** (largest

positive contribution), together with *purpose* and *dti*; offsetting factors include *open_acc* and *total_acc*. These patterns align with economic intuition and suggest policy levers: **first**, tighten or reprice for profiles with **high interest rate + high revolving utilization + long term**; **second**, prioritize income/credit verification (since *Verified* lowers risk); **finally**, use local SHAP explanations to justify borderline decisions in manual review.

4.2.5. Business implementation

Model choice must align with business objectives. The comparison table shows that no single model is universally best.

Table 3: Business Tradeoff

Objective ▼	Metric Focus ▼	Best Model ▼
Minimise financial losses (catch all defaults)	Recall	Logistic Regression
Maximise loan approval revenue while controlling losses	F1 / AUC (balanced view)	XGBoost
Explain decisions to regulators	Interpretability	Logistic Regression
Automate large-scale screening	Speed + Stability	XGBoost / LightGBM

Logistic Regression delivers strong recall and full interpretability. It is therefore appropriate when the institution must justify individual lending decisions to regulators and senior risk committees. LightGBM is efficient for large scale monitoring and portfolio stress testing. However, for production credit decisions on new applications, the tuned XGBoost model is preferred.

Although Logistic Regression and LightGBM achieve recall that is only about two percentage points higher, this improvement comes at the cost of lower precision, which would generate more false positives. In practice, the cost of excessive false positives is operational. It increases manual review load, reduces approval volume, and may block borrowers who would have repaid. The tuned XGBoost model provides the best balance between recall, precision, and AUC, and produces a stable probability ranking that supports pricing and approval strategy.

The model output can be then converted into a **four-tier risk segmentation policy** using predicted default probability.

Table 4: Risk Segmentation Policy

Risk Tier	Probability Range	Business Action
Low Risk	< 0.25	Auto-approve
Medium Risk	0.25 – 0.50	Manual review
High Risk	0.50 – 0.75	Approve with higher interest or collateral
Very High Risk	≥ 0.75	Reject application

Low risk applicants, probability below 0.25, can be auto approved. Medium risk applicants, probability between 0.25 and 0.50, are routed to analyst review. High risk applicants, probability between 0.50 and 0.75, may still be funded, but only with tighter terms such as higher interest rate or collateral. Very high-risk applicants, probability at or above 0.75, are declined. This structure links model output directly to financial action and allows the lender to tune its risk appetite by adjusting tier thresholds.

The final recommended pipeline is therefore: use tuned XGBoost as the core scoring engine for real time lending decisions, apply the four-tier probability segmentation to operationalise approval and pricing, and retain Logistic Regression in parallel to provide an interpretable benchmark for regulatory explanation and model governance.

5. Discussion and conclusions

We built an origination-time default-risk model for LendingClub using only pre-funding variables and time-aware validation (train-2016/val-2017/test-2018). Across Logistic Regression, LightGBM, and XGBoost, performance is consistent; the tuned XGBoost is our champion (test ROC-AUC $\approx$ 0.708, PR-AUC $\approx$ 0.306, recall $\approx$ 0.67). To interpret the champion at both global and case levels, we computed SHAP values and grouped them back to raw features to avoid one-hot fragmentation. The patterns are economically intuitive: **int_rate** is the dominant risk driver, while higher **revol_util**, **term (60 months)**, **loan_amnt**, **dti**, and **revol_bal** push risk upward; in contrast, stronger capacity/quality

signals-higher **annual_inc**, deeper credit depth (**open_acc**, **total_acc**), better **grade**, **home_ownership = MORTGAGE**, and **verification_status = Verified** reduce risk. Local SHAP explanations show that high-risk applicants are typically propelled by the combination of high interest rate, high revolving utilization, and long term, whereas depth of credit and verified information often offset risk. These insights align with our four-tier policy (auto-approve, manual review, risk-priced, decline) and clarify the levers behind individual scores.

Several limitations apply. Results are conditioned in 2016–2018 vintages, so macro shifts and policy drift can change base rates and feature distributions; out-of-time performance may move without periodic refits and monitoring. The training data includes only accepted loans, which introduces selection bias for application-stage deployment. While ranking is strong, probability calibration for pricing/provisioning was not our primary focus, and features are predominantly borrower-level (absent explicit macro indicators), so the model may under-react in stress regimes. Finally, SHAP explains marginal model effects rather than strict causality; grouping one-hot levels back to raw features improves readability but can smooth heterogeneity across categories.

Looking forward, we outline practical extensions. **First**, calibrate PDs (isotonic or Platt) and set operating thresholds by expected loss or portfolio utility, then back-test within the four-tier workflow. **Second**, make the system macro-aware (e.g., unemployment and rate environment) and institute PSI/score-drift monitors with scheduled refits in a champion–challenger setup (retain LightGBM as challenger and promote it if it consistently wins at the business operating point). **Finally**, build on what is already in place: we have integrated coefficient-based LR for transparency and both global (grouped) and local SHAP explanations for reviewers; the next governance steps are to maintain a model card, add fairness/stability diagnostics across key segments, and document reviewer playbooks that use SHAP for borderline decisions.

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APPENDIX A: Google Colab Code

All data analysis and machine learning experiments for the LendingClub loan default prediction project were conducted using **Google Colab**, an online Python-based notebook environment that supports GPU and cloud-based computation. Colab was selected for its accessibility, reproducibility, and seamless integration with Python data science libraries.

Google Colab link:

<https://drive.google.com/file/d/1koR45lw0lwEAXX1F1aEYyqzj4OPH4Yd-/view?usp=sharing>

APPENDIX B: Data Source

Dataset is collected from Kaggle: [All Lending Club loan data](#)

Account Activity	Borrower Demographics	Co-Applicant	Credit History & Delinquency	Financial Ratios & Balances	Loan Characteristics & Terms
acc_now_delinq acc_open_past_24mths chargeoff_within_12_mths collections_12_mths_ex_med mort_acc mths_since_recent_bc mths_since_recent_bc_dlq mths_since_recent_inq mths_since_recent_revol_delinq open_acc open_acc_6m open_act_il open_il_12m open_il_24m open_rv_12m open_rv_24m	annual_inc annual_inc_joint application_type emp_length emp_title home_ownership verification_status verified_status_joint Collapse ^	sec_app_earliest_cr_line sec_app fico_range_high sec_app fico_range_low sec_app_inq_last_6mths sec_app_mort_acc sec_app_num_rev_accts sec_app_open_acc sec_app_open_act_il sec_app_revol_util Collapse ^	delinq_2yrs inq_fl inq_last_12m inq_last_6mths mths_since_last_delinq mths_since_last_major_derog mths_since_last_record num_accts_ever_120_pd num_actv_bc_tl num_actv_rev_tl num_bc_sats num_bc_tl num_il_tl num_op_rev_tl num_rev_accts num_rev_tl_bal_gt_0 num_sats num_tl_120dpd_2m num_tl_30dpd num_tl_90g_dpd_24m pub_rec pub_rec_bankruptcies tax_liens Collapse ^	dli dli_joint tot_coll_amt tot_coll_amt tot_cur_bal tot_hi_cred_lim tot_hi_cred_lim_1 total_acc total_bal_ex_mort total_bal_il total_bc_limit total_cu_tl total_il_high_credit_limit total_rev_hi_lim Collapse ^	grade initial_list_status installment int_rate issue_d loan_amnt purpose sub_grade term Collapse ^ Settlement & Hardship debt_settlement_flag disbursement_method hardship_flag settlement_amount settlement_date settlement_percentage settlement_status settlement_term Collapse ^

Figure 28: Data Overview

APPENDIX C: Data Dictionary

Data dictionary for 26 features (for running model):

Feature	Type	Short description	Notes
issue_d	Date	Loan issue date	Use for time-based split; exclude from training
issue_year	Int	Issue year	Use for split; exclude from training
loan_default_flag	Target (0/1)	Default label	1 = Charged Off/Default, 0 = Fully Paid
loan_amnt	Numeric	Loan amount (USD)	—
term	Categorical	Loan term	Usually “36 months” or “60 months”
int_rate	Numeric (%)	Nominal interest rate	—
installment	Numeric	Monthly payment (USD)	Function of loan_amnt, term, int_rate
grade	Categorical	LC-assigned risk grade	A–G (A = best)
purpose	Categorical	Loan purpose	e.g., debt_consolidation, credit_card, car
initial_list_status	Categorical	Initial listing type	w = whole loan, f = fractional
annual_inc	Numeric	Self-reported annual income (USD)	—
home_ownership	Categorical	Home ownership status	RENT/MORTGAGE/OWN/OTHER
verification_status	Categorical	Income verification status	Verified / Source Verified / Not Verified
dti	Numeric (%)	Debt-to-income ratio	Excludes the current loan
fico_range_low	Numeric	Lower bound of FICO score	—
fico_range_high	Numeric	Upper bound of FICO score	—
open_acc	Numeric (count)	Number of open credit lines	—
total_acc	Numeric (count)	Total credit accounts on file	—
revol_bal	Numeric	Revolving balance outstanding (USD)	—
revol_util	Numeric (%)	Revolving utilization rate	Typically 0–100(+)
delinq_2yrs	Numeric (count)	Delinquencies in past 2 years	—
inq_last_6mths	Numeric (count)	Credit inquiries in last 6 months	—
pub_rec	Numeric (count)	Public derogatory records	Judgments, liens, etc.
pub_rec_bankruptcies	Numeric (count)	Bankruptcy filings	—
collections_12_mths_ex_med	Numeric (count)	Collection events in last 12 months (ex. medical)	—
chargeoff_within_12_mths	Numeric (count)	Charge-offs within last 12 months	—

Note: issue_d and issue_year are used only for chronological splitting and should be dropped from the training features to avoid time leakage. All other variables are application-time features.

APPENDIX D: Use of ChatGPT in Report Development

ChatGPT was used as a **writing and refinement tool** throughout the development of this report to support academic clarity, consistency, and professionalism in communication. The authors utilised ChatGPT to:

- **Polish and refine written content** – improving sentence structure, conciseness, and coherence across technical sections such as data preprocessing, methodology, results interpretation, and discussion.
- **Ensure academic tone and readability** – rewriting initial drafts into formal, research-appropriate language while maintaining the original meaning and analytical depth.
- **Summarise and synthesise complex findings** – generating concise summaries for empirical results, reflection journals, and project overviews.
- **Assist with structure and formatting** – suggesting outlines for key sections (e.g., methodology framework, cost-benefit analysis) and improving logical flow between subsections.
- **Generate supporting material** – producing consistent reflection journal entries, abstracts, and appendices aligned with the project’s academic requirements.

No model-generated content was accepted without **critical review and manual revision** by the authors. All analytical insights, interpretations, and coding processes remain the original work of the project team. ChatGPT’s role was limited to linguistic enhancement, clarity improvement, and report presentation support.

APPENDIX E: Logistic Regression tuning

```
# 1) Tunning
n = X_tr.shape[0]
m = min(100_000, n)
idx = np.random.RandomState(42).choice(n, size=m, replace=False)
X_tr_tune = X_tr[idx]
y_tr_tune = y_train.iloc[idx] if hasattr(y_train, "iloc") else y_train[idx]

Cs = [0.05, 0.1, 0.3, 1, 3]
best_pr, best_C = -1, None
```

```
Best C = 0.05 (val PR-AUC=0.3999)
Best threshold (val) = 0.517 (F1=0.460)

=== Test metrics (tuned LR) ===
ROC-AUC : 0.7011
PR-AUC   : 0.2889
LogLoss  : 0.6263
Accuracy: 0.6504
Precision: 0.2552
Recall   : 0.6354
F1       : 0.3642
Confusion matrix (test):
[[30867 16387]
 [ 3223  5616]]
Classification report (test):
```

	precision	recall	f1-score	support
0	0.91	0.65	0.76	47254
1	0.26	0.64	0.36	8839
accuracy			0.65	56093
macro avg	0.58	0.64	0.56	56093
weighted avg	0.80	0.65	0.70	56093

We tuned LR in a lean, fast manner: fixed L2 penalty, SAGA solver, and `class_weight="balanced"`, and searched only $C \in \{0.05, 0.1, 0.3, 1, 3\}$. To reduce runtime, the search was run on a 100k random subsample of the training set and ranked by PR-AUC on the validation set (appropriate for class imbalance). The best value was $C = 0.05$ (val PR-AUC ≈ 0.400)

APPENDIX F: Contribution table

Member	Tasks
Trang Linh Tran – 25597713	– Timeline organisation, task distribution, participation in writing methodology, data processing and result discussion, executive summary and appendices, maintaining coherence and correct formatting for reports.
Thi Thu Phuong Nguyen – 25536893	Implement feature encoding, build/run the models, perform hyperparameter tuning, and coordinate with the reporting team to contribute to the report.
Thai Minh Huong Dinh - 25504554	- Code the exploratory data analysis, data preprocessing, splitting the data, collaborate with the reporting team to contribute to the write-up and resolve any issues.
Raymond Chang - 25618810	- Train and evaluate models, tune hyperparameters, and work with the reporting team on the write-up.
Thi Nhat Phuong Bui – 25609138	– Introduction, Project Objectives, Data Understanding, EDA report
Yining Wang - 25605369	Project Scope, Experiment Plan